Date: 5/14/2025
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Lab 6: Indentation
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Overview

Experimental Setup

Figure 1 shows the experimental setup:

- Criterion Model 43 MTS Universal Test System with a 30 kN load cell
- IPhone
- Tripod
- Samples:
    - ➔ ABS uniaxial tension sample (ASTM D638-14, Type I)
    - ➔ ABS notched sample (same material with a notch)
    - ➔ TPU notched sample
    - ➔ All samples are 3D-printed, then spray-painted with a random speckle pattern for DIC

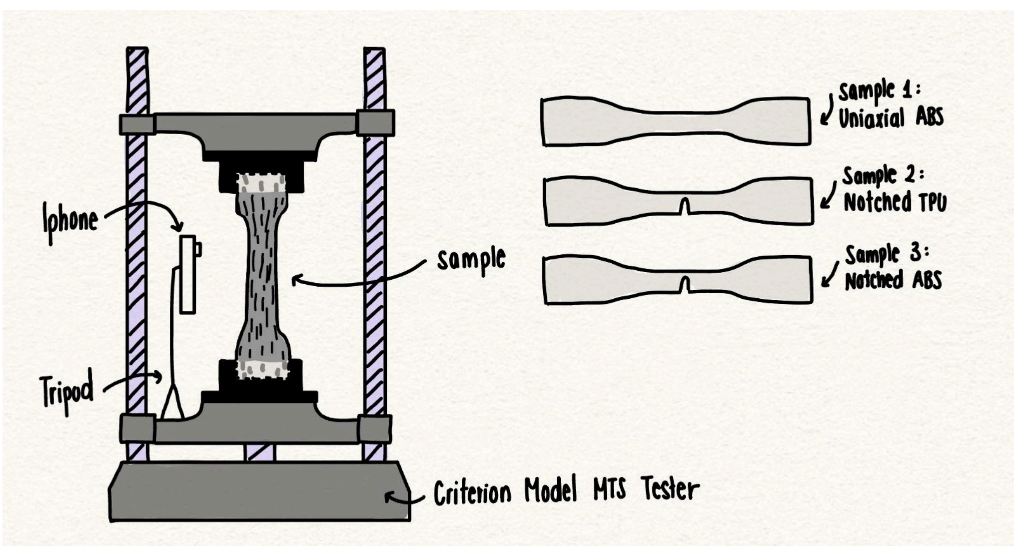

Figure 1 shows the initial dimensions and test setup for all samples. Before testing, we measured each sample's narrow-section width, notch width, thickness, and gauge length. Every sample had a narrow width of 13 mm, a notch width of 6.5 mm, a thickness of 2.8 mm, and a gauge length of 70 mm. We then mounted the samples in the servo-hydraulic MTS tester and applied a constant strain rate of 2×10^{-3} s^{-1} (0.14 mm/s) for ABS and 2×10^{-2} s^{-1} (1.4 mm/s) for TPU until failure. Throughout each test, we recorded load–displacement data and captured video of the entire tensile process for use in Lab 7.

Raw Data

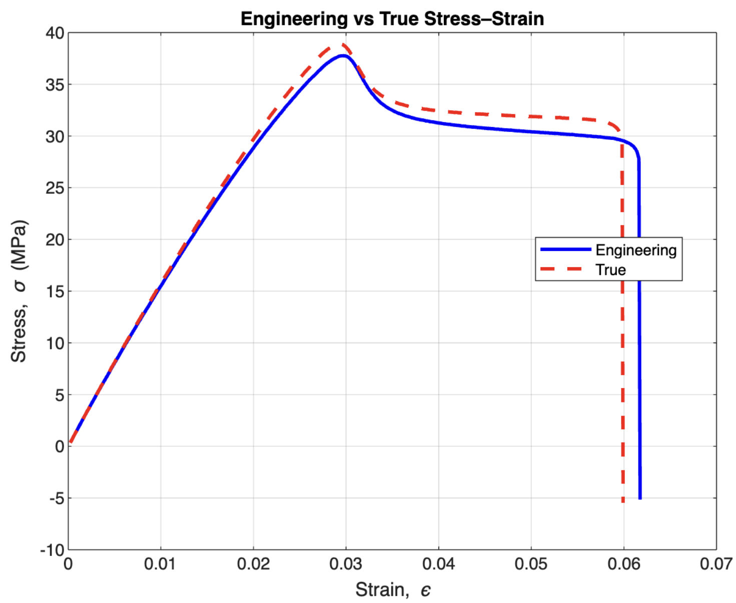

Figure 2 shows both the engineering and true stress–strain curves for the uniaxial ABS sample (Type I, ASTM D638-14). True strain was determined as the natural logarithm of (1 + engineering strain), and true stress was obtained by multiplying the engineering stress by (1 + engineering strain). All tests were performed at a constant rate of 0.14 mm/s. As expected, the true stress curve rises above the engineering curve due to the continuous reduction in cross-sectional area during deformation. However, the sample broke soon after necking, so it couldn't reach the higher true stress values that would come afterward.

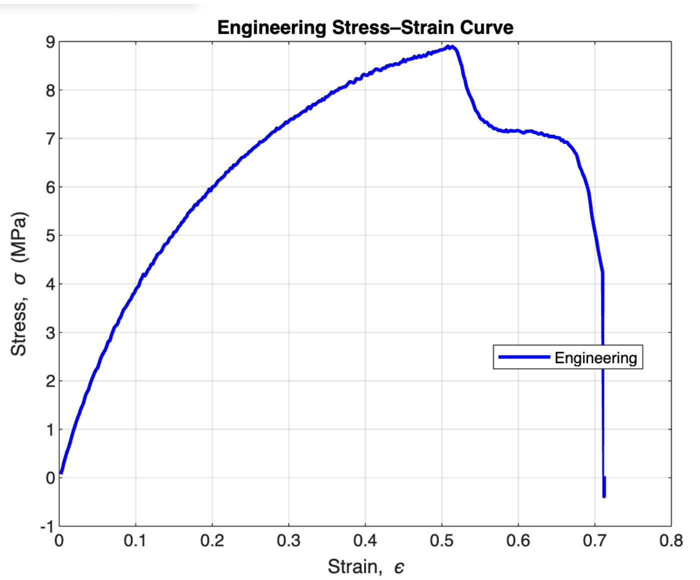

Figure 3 shows the engineering stress–strain curve for the notched TPU sample (Type I, ASTM D638-14) tested at a rate of 1.4 mm/s (notch width = 6.5 mm). The material shows an initial linear elastic region up to a peak stress of ≈8.8 MPa at a strain of ≈0.55, followed by strain softening and final fracture at ≈0.72 strain.

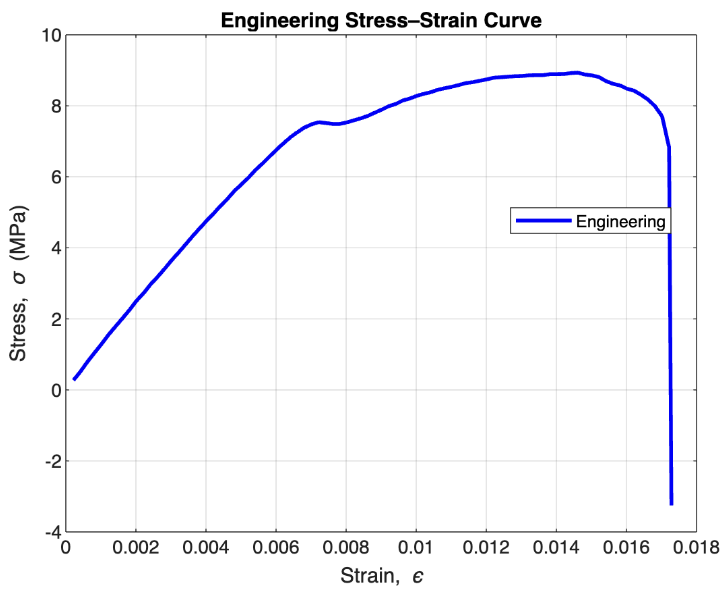

Figure 4 shows the engineering stress–strain curve for the notched ABS specimen (Type I, ASTM D638-14) tested at a rate of 0.14 mm/s (notch width = 6.5 mm). The curve shows a linear elastic region up to yield, followed by strain hardening to a peak stress of about 9 MPa at around 1.6% strain, then a sudden drop indicating fracture.

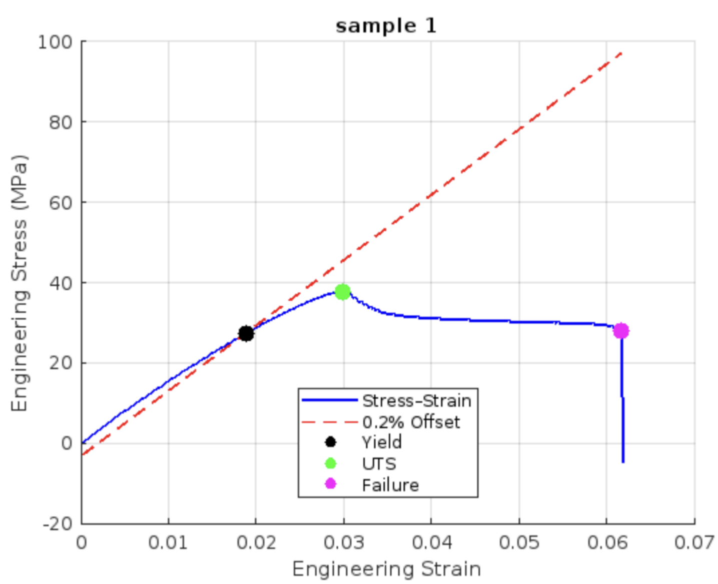

Figure 5 shows the engineering stress–strain curve for the uniaxial ABS sample (Type I, ASTM D638-14) tested at 0.14 mm/s. The initial slope gives an elastic modulus of about 1.63 GPa. The red line is the 0.2 % offset line used to identify the yield strength of 27.4 MPa.. The ultimate tensile strength is 37.8MPa, and failure occurs at 27.8 MPa with a strain of 0.0616 (6.16 % elongation).

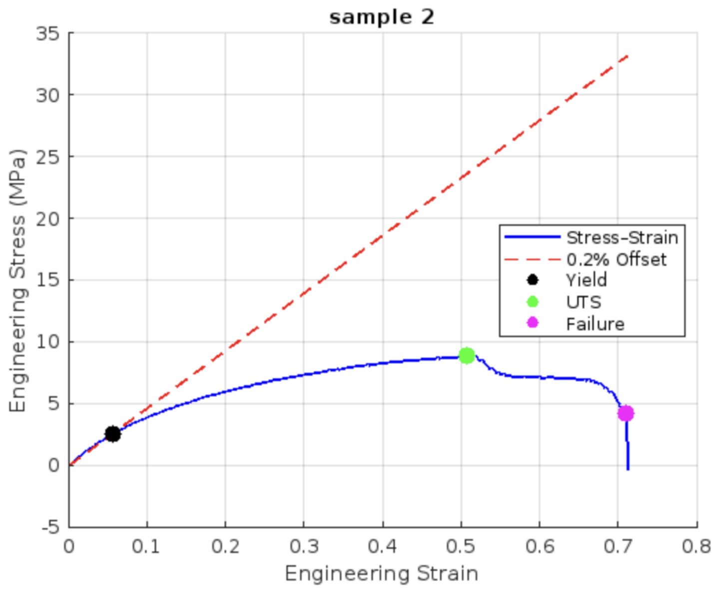

Figure 6 shows the engineering stress–strain curve for the notched TPU sample (Type I, ASTM D638-14) tested at 0.14 mm/s (notch width = 6.5 mm). The red dashed line is the 0.2 % offset used to find the yield strength (2.53 MPa at $\varepsilon \approx 0.0561$), the green marker shows the ultimate tensile strength (8.91 MPa at $\varepsilon \approx 0.5081$), and the magenta marker indicates final failure (4.24 MPa at $\varepsilon \approx 0.7101$, 71.01 % elongation).

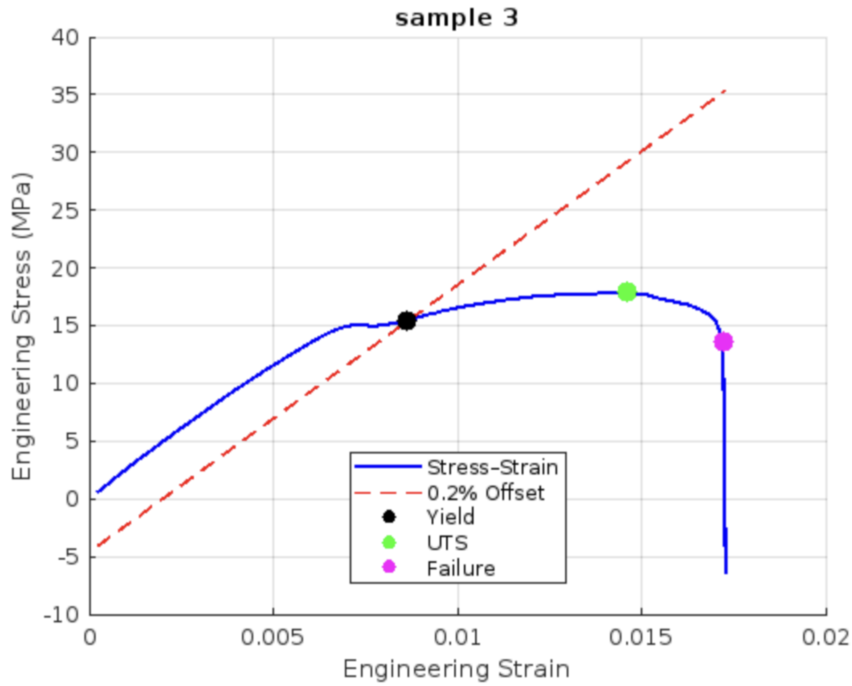

Figure 7 shows the engineering stress–strain curve for the sample (Type I, ASTM D638-14) tested at 0.14 mm/s (notch width = 6.5 mm). The red dashed line is the 0.2 % offset used to find the yield strength of 15.43 MPa (ε≈0.0086). The initial slope gives an elastic modulus of about 2.31 GPa. The ultimate tensile strength of 17.87 MPa occurs at ε≈0.0146, and failure happens at 13.66 MPa with a strain of 0.0172 (1.72 % elongation).

Table 1. Material properties for the 3 samples

	Elastic modulus (MPa)	**0.2% yield strength (MPa)**	**Ultimate tensile strength (MPa)**	**Failure strength (MPa)**	**Elongation at failure (%)**
Sample 1: uniaxial ABS	**1628**	**27.4**	**37.8**	**27.8**	**6.16**
Sample 2:notched TPU	**46.71**	**2.53**	**8.91**	**4.24**	**71.01**
Sample 3: notched ABS	**2314.13**	**15.43**	**17.87**	**13.66**	**1.72**

When comparing the elastic modulus of the ABS uniaxial testing sample and the elastic modulus for ABS from the indentation test (Lab 5), we notice that the uniaxial tensile test yielded an elastic modulus of 1,628 MPa for ABS, whereas the indentation test in Lab 5 produced a higher value of 2.82 GPa. The tensile measurement is considered more reliable because it evaluates the material's elastic behavior along a consistent gauge length and cross-section. Indentation testing only measures a small area and relies on hardness-to-modulus correlations, which can introduce additional sources of error.

Using the Tabor relation $H = 3^* \sigma_{yield}$, the Lab 5 hardness of 0.1579 GPa predicts a yield strength of about 52.6 MPa. By contrast, our uniaxial tensile test gave a yield strength of only 27.4 MPa. The direct tension test is more accurate because it measures the material's actual response to pulling, whereas the Tabor factor is a rough shortcut that can overestimate yield strength when you convert from hardness.

Figure 8 shows that the uniaxial TPU sample withstood being stretched to the MTS tester's maximum extension without breaking.